

# ENGR. ABDUL MURAD

**Mechanical Engineer | PEC Registered Engineer | Master's & Scholarship Applicant**

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## ACADEMIC PROFILE

PEC-registered Mechanical Engineer with professional experience across HVAC/MEP engineering, Revit BIM coordination, QA/QC, equipment support and technical documentation. Academic interests center on sustainable energy systems, HVAC energy efficiency, solar-assisted thermal processes, renewable-energy applications and engineering design. Seeking an internationally funded Master's or research-based graduate opportunity where practical engineering experience can support advanced study and applied research.

## EDUCATION

**BSc Mechanical Engineering | Hamdard University, Karachi, Pakistan**

- Core areas: engineering mechanics, thermodynamics, fluid mechanics, heat transfer, machine design, manufacturing, materials, CAD, controls, HVAC/refrigeration, energy systems, engineering economics and project management.
- Medium of instruction: English; English proficiency certificate available upon request.

**FSc Pre-Engineering | Faizan School & College, Besham, Shangla KPK**

## RESEARCH & ACADEMIC PROJECTS

**Solar-Powered Desalination with Forced-Air Bubbling | Final Year Research**

- Integrated design, mathematical modelling, fabrication, testing and performance evaluation of a solar-powered desalination concept using forced-air bubbling.
- Prepared an academic monograph and a journal-format research manuscript from the project work.

**Solar Tracker System | Semester Project**

- Developed a controlled solar-panel orientation concept to improve exposure to available sunlight.

**Smart Solar Home System | Semester Project**

- Developed a small residential concept integrating solar power, household loads and energy-aware control.

**Energy Storage System Using Supercapacitor (ESSC) | Semester Project**

- Explored fast-charge energy-storage concepts using a supercapacitor-based arrangement with basic monitoring and load demonstration.

**Research Design & CAD Development | Technical Project**

- Produced technical drawing and design-development support for the experimental desalination system.

## PROFESSIONAL EXPERIENCE

**MEP Engineer Mechanical | Pro Building Designer, Pakistan**

**Aug 2026 - Present**

- MEP & HVAC engineering; Revit BIM modelling; architectural/MEP coordination; HVAC ductwork and air-distribution documentation.
- Existing-building documentation, renovation/as-built updates, scan-to-BIM support, clash coordination, Revit families/views/schedules and MEP QA/QC.

**Mechanical Engineer (HVAC) | Orient Energy Pvt. Ltd., Karachi, Pakistan**

**Jul 2025 - May 2026**

- Planned and tracked inspection/repair activities, coordinated work orders, supported troubleshooting, parts replacement and maintenance reporting.

**Mechanical Engineer | Shahnawaz Engineering Pvt. Ltd., Karachi, Pakistan**

**Aug 2024 - Apr 2025**

- Monitored temperature, pressure and flow; supported chilled-water operation, water treatment, routine maintenance, troubleshooting, energy-efficiency checks and safety compliance.

ADDITIONAL ENGINEERING EXPERIENCE

- Application Engineer | Cosmix Engineering, Karachi, PakistanJun 2023 - Sep 2023
- Provided HVAC technical support and equipment selection for chillers, AHUs, FCUs and ducted systems.
  - Performed load calculations, duct layout/sizing and air-distribution work; prepared shop drawings, BOQs, technical submittals and proposals; supported site installation, commissioning and performance testing.
- Mechanical Engineer | Pakistan Railway, PakistanJul 2022 - May 2023
- Supported equipment inspection and condition monitoring, maintenance procedures, technical documentation, troubleshooting coordination and spare-parts tracking.

TECHNICAL SKILLS

|                           |                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------|
| BIM / CAD                 | Revit MEP, AutoCAD MEP, BIM 360, SolidWorks                                                       |
| HVAC / MEP                | Load calculations, duct layout & sizing, air distribution, piping/MEP coordination                |
| Engineering Documentation | Shop drawings, BOQs, MTO, technical submittals, as-built updates, schedules                       |
| QA/QC & Standards         | Inspection documentation, material traceability, pressure testing, ISO/ASME awareness             |
| Research & Analysis       | Experimental planning, mathematical modelling, testing, performance evaluation, technical writing |
| Productivity              | Microsoft Word, Excel, Visio and PowerPoint                                                       |

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Pakistan Engineering Council - Registered Engineer (Registration No. 059962), 2025
- IEEE Certificate of Appreciation, 2024
- ISO 45001:2018 Occupational Health & Safety Management System Awareness, 2025
- COSHH Risk Assessor Certification, 2025
- ISO 50001:2018 Energy Management System Awareness, 2025
- ISO 14001:2015 Environmental Management System Awareness, 2025
- AutoCAD Training Certificate - Virtual University Islamabad, 2024
- Project Management Certificate - Great Learning, 2024
- Certificate in Information Technology - Hi-Tech Institute of Technology, 2015

MEMBERSHIPS & PROFESSIONAL AFFILIATIONS

- Registered Member, IAENG - International Association of Engineers (2026)
- ASME Membership (2021 - Present)
- ASME Chapter, Hamdard University Karachi (2022)
- Registered Mechanical Engineer, Pakistan Engineering Council (2025)

SELECTED CONFERENCES, SEMINARS & WORKSHOPS

- Academic and Industrial Applications of CFD - ASME-HU Chapter
- Hands-on Training on DS SOLIDWORKS - ASME-HU Chapter
- Two-Day Session on Web 3.0 - Hamdard University
- International Scholarship Opportunities in Higher Education - ASHRAE & EMBS Student Chapters, UET Lahore
- International Education & Entrepreneurship - MUST / Kamyab Jawan Markaz
- Sustainable Energy Solutions for Future - NED University, Mechanical Engineering

LANGUAGES

English - Fluent (medium of instruction for BSc degree; proficiency certificate available upon request) | Urdu - Native

Graduate Study Interests: Sustainable Energy Systems | HVAC & Building Energy | Thermal/Fluid Systems | Renewable Energy Applications | BIM-enabled MEP Engineering | Engineering Design & Applied Research